

Categorical Homotopy Theory - Self Study

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Abstract

This text contains my notes on Categorical Homotopy Theory, by Emily Riehl. I read this book independently during the summer of 2025, between my sophomore and junior years. My notes largely paraphrase the text. Going into the text, I lacked a precise understanding of model category theory, and had a weak intuition for simplicial sets/objects and methods. I also lacked some helpful algebraic topology knowledge. I continuously supplemented my readings with references to Hatcher's *Algebraic Topology*, Hovey's *Model Category Theory*, and on occasion May's *Simplicial Objects in Algebraic Topology*. I intended to, as a side effect of reading this text, improve my understanding of these subjects such that I might only need to skim texts on them in the future, rather than read them in their entirety.

Notation. Disjoint unions are denoted by $\amalg$. Composition of functors and morphisms is denoted by $\circ$, or juxtaposition when the notation can be well understood. Vertical composition of 2-cells is denoted by $\circ$, and horizontal composition is denoted by $\cdot$. Usually, juxtaposition of 2-cells denotes horizontal composition, but on occasion it may denote vertical composition if it can be well understood that there is no horizontal composition. For a category $\mathcal{C}$, I also write $\mathcal{C}$ to denote the set of objects of $\mathcal{C}$. On occasion, I will instead write $\text{Obj}(\mathcal{C})$ for emphasis. I write $\text{Mor}(\mathcal{C})$ for the set of arrows of $\mathcal{C}$.

Chapter 1

All concepts are Kan extensions

I already know Kan extensions so I will breeze through this chapter.

1.1 Kan Extensions

Definition 1.1 (Lan Extension). Given functors $F : \mathcal{C} \rightarrow \mathcal{E}$ and $K : \mathcal{C} \rightarrow \mathcal{D}$, a *left Kan extension* of F along K is a functor $\text{Lan}_K F : \mathcal{D} \rightarrow \mathcal{E}$ and a natural transformation $\eta : F \Rightarrow \text{Lan}_K F \circ K$ (a pasting diagram) which is initial with respect to this property; for any other extension $G : \mathcal{D} \rightarrow \mathcal{E}$ and $\gamma : F \Rightarrow G \circ K$, there is a unique natural transformation $\nu : \text{Lan}_K F \Rightarrow G$ such that $\gamma \cdot 1_K = (\nu \cdot 1_F) \circ \eta$. Dually, a right Kan extension is terminal with respect to natural transformations $\varepsilon : \text{Ran}_K F \circ K \Rightarrow F$. ♥

Remark. These definitions generalize to any 2-category. We usually specialize in the 2-category $\mathcal{Cat}$ for obvious reasons, so we reserve the term “Kan extension” for $\mathcal{Cat}$. When working in $\mathcal{Cat}$, I use “extension” to mean a pair (G, γ) satisfying the extension property, but not necessarily universally.

In unenriched category theory, universal properties (in this text) are representations of functors. A left Kan extension $\text{Lan}_K F$ with notation as above is a representation for

$$\mathcal{E}^{\mathcal{C}}(F, - \circ K) : \mathcal{E}^{\mathcal{C}} \rightarrow \mathbf{Set}.$$

To justify this, recall here that universality gives for any G ,

$$\mathcal{E}^{\mathcal{C}}(F, G \circ K) \cong \mathcal{E}^{\mathcal{D}}(\text{Lan}_K F, G).$$

Hence, $\text{Lan}_K F$ is a representation as the universal object. Alternatively, the Yoneda lemma promotes any extension $\gamma : F \Rightarrow G \circ K$ to a natural transformation

$$\gamma : \mathcal{E}^{\mathcal{C}}(G \circ K, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, -).$$

Precomposing by $1 \times K$ gives

$$\gamma \cdot 1 : \mathcal{E}^{\mathcal{C}}(G \circ K, - \circ K) = \mathcal{E}^{\mathcal{D}}(G, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, - \circ K).$$

Universality is then equivalent to that the natural transformation

$$\eta : \mathcal{E}^{\mathcal{D}}(\text{Lan}_K F, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, - \circ K)$$

be an isomorphism.

By universality, $\text{Lan}_K -$ is functorial provided it always exists, so this gives an adjunction for precomposition K^* :

$$\mathcal{E}^{\mathcal{D}}(\text{Lan}_K F, G) \cong \mathcal{E}^{\mathcal{C}}(F, G \circ K) \quad \text{Lan}_K \dashv K^* \dashv \text{Ran}_K.$$

1.2 A formula

Theorem 1.1. *Let $\mathcal{C}$ small, $\mathcal{D}$ locally small, and $\mathcal{E}$ cocomplete. Then any $F : \mathcal{C} \rightarrow \mathcal{E}$ has an extension along any $K : \mathcal{C} \rightarrow \mathcal{D}$ given pointwise by*

$$\mathrm{Lan}_K F(d) = \int^{c \in \mathcal{C}} \mathcal{D}(K(c), d) \cdot F(c)$$

for $\cdot$ the copower/tensor and $d \in D$. Dually,

$$\mathrm{Ran}_K F(d) = \int_{c \in \mathcal{C}} F(c)^{\mathcal{D}(K(c), d)}.$$

Here, exponentiation is the power/cotensor.

| **Proof.** See my notes for Mac Lane. ■

Theorem 1.2. *With notation as in 1.1, we also have*

$$\mathrm{Lan}_K F(d) = \varinjlim \left(F \circ \Pi^d \right), \quad \Pi^d : (K \downarrow d) \rightarrow \mathcal{C},$$

for Π^d the comma category projection. We will prove these formulas again in the context of weighted limits later. Dually,

$$\mathrm{Ran}_K F(d) = \varprojlim \left((d \downarrow K) \xrightarrow{\Pi_d} \mathcal{C} \xrightarrow{F} \mathcal{E} \right).$$

1.3 Pointwise Kan extensions

Definition 1.2 (Preserves). A functor $L : \mathcal{E} \rightarrow \mathcal{F}$ preserves left Kan extensions if $(L \mathrm{Lan}_K F, L\eta)$ is the left Kan extension $\mathrm{Lan}_K LF$. ♥

The following holds even when coends and colimits don't exist. Again, see my notes on Mac Lane for the (simple) proof.

Lemma 1.1. *Left adjoints preserve left Kan extensions. Right adjoints preserve right Kan extensions.*

It has generally been decided that Kan extensions in full generality are not nearly as important as the following special case, and the accompanying theorem which reconciles the terminology.

Definition 1.3 (Pointwise Kan extensions). If $\mathcal{E}$ is locally small, a pointwise right Kan extension is one preserved by all representable functors $\mathcal{E}(e, -)$. By contravariance, a pointwise left Kan extension is one such that postcomposition by $\mathcal{E}(-, e)$ with the opposite (right) Kan extension gives a right Kan extension. ♥

Theorem 1.3. *A right Kan extension is pointwise if and only if it is computed by limits as described in theorem 1.2. A left Kan extension is pointwise if and only if it is computed by colimits as in theorem 1.2.*

1.4 All concepts

The notion of Kan extensions subsumes all the other fundamental concepts of category theory.

Theorem 1.4. *Let $F : \mathcal{C} \rightarrow \mathcal{E}$. A limit of F is precisely a right Kan extension of F along the terminal functor $! : \mathcal{C} \rightarrow \mathbf{1}$. A left Kan extension along the terminal functor is precisely a colimit.*

Theorem 1.5. *Adjunctions $(F, G, \eta, \varepsilon)$ are precisely left Kan extensions (G, η) of $\mathbf{1}$ along F such that F preserves the Kan extension.*

Kan extensions also give the Yoneda lemma. For $F : \mathcal{C} \rightarrow \mathbf{Set}$,

$$F(c) \cong \text{Ran}_{\mathbf{1}_e} F(c) \cong \int_{x \in \mathcal{C}} F(x)^{\mathcal{C}(c, x)} \cong \int_{x \in \mathcal{C}} \mathbf{Set}(\mathcal{C}(c, x), F(x)) \cong \mathbf{Set}^e(\mathcal{C}(c, -), F).$$

The first isomorphism is kind of just obvious, the second is the pointwise construction via ends, the third is definition of exponentials in $\mathbf{Set}$, and the fourth is an identity I proved in my notes on Mac Lane. This also gives us the coYoneda lemma:

$$F(c) \cong \int^{x \in \mathcal{C}} \mathcal{C}(x, c) \cdot F(x).$$

Finally, if $\mathcal{E}$ is complete and K is fully faithful, then the counit is an isomorphism $\text{Ran}_K F \circ K \cong F$.

1.5 Adjunctions involving simplicial sets

We denote by $\mathbf{sSet}$ the functor category $\text{Fun}(\Delta^{\text{op}}, \mathbf{Set})$, where Δ is the simplex category, and Δ_+ is the augmented simplex category which includes $[-1]$. $\mathbf{sSet}$ is the free colimit completion of Δ , although this is not unique to Δ and holds for any small category.

Let $\mathcal{E}$ be cocomplete and locally small, and let $\Delta^\bullet : \Delta \rightarrow \mathcal{E}$ be covariant. Write Δ^n for the image $\Delta^\bullet[n]$; similarly for $X \in \mathbf{sSet}$, write X_n for the set $X[n]$ of n -simplices and $X_\bullet : \Delta^{\text{op}} \rightarrow \mathbf{Set}$ when we wish to emphasize its functorial role, rather than its role as an assignment of sets. Define $L = \text{Lan}_y \Delta^\bullet$ as the left Kan extension of $\Delta^\bullet$ along the Yoneda embedding y :

$$\begin{array}{ccc} \Delta & \xrightarrow{\Delta^\bullet} & \mathcal{E} \\ & \searrow y & \downarrow \cong \\ & & \mathbf{sSet} \end{array} \quad \begin{array}{c} \nearrow L \\ \end{array}$$

Since the Yoneda embedding is fully faithful and $\mathcal{E}$ is locally small and cocomplete, the unit is an isomorphism by the dual of the above statement. The functor L is then defined via theorem 1.1 as

$$L(X) := \int^{n \in \Delta} \mathbf{sSet}(y(n), X) \cdot \Delta^n \cong \int^{n \in \Delta} X_n \cdot \Delta^n.$$

The first isomorphism is once again theorem 1.1, and the second is the Yoneda lemma since $y(n) = \Delta(-, n)$ and indeed $\text{Nat}(\Delta(-, n), X) \cong X(n) = X_n$. Since coends are colimits, which are characterized by coproducts and coequalizers, we obtain an explicit description as

$$L(X) = \text{coeq} \left(\coprod_{f \in \text{Mor}(\Delta)} X_{\text{cod } f} \cdot \Delta^{\text{dom } f} \rightrightarrows \coprod_{[n] \in \Delta} X_n \cdot \Delta^n \right).$$

The definition of L on the arrows is given by universality. We usually choose to denote by Δ^n the simplicial set $y[n]$, and by cocompleteness we have

$$(L \circ y)[n] = \Delta^n,$$

somewhat justifying this different notation we have used here.

Since colimits commute, L preserves colimits, so the adjoint functor theorem gives an adjoint $R : \mathcal{E} \rightarrow \mathbf{sSet}$, which has for any $e \in \mathcal{E}$,

$$R(e)_n \cong \mathbf{sSet}(y[n], R(e)) \cong \mathcal{E}(Ly[n], e) \cong \mathcal{E}(\Delta^n, e).$$

The first isomorphism is Yoneda, the second is the adjunction relation, and the third is our above discussion. Thus, n -simplices of $R(e)$ can be defined as the set of maps from $\Delta^n \rightarrow e$ in $\mathcal{E}$, making $R = \mathcal{E}(\Delta^\bullet, -)$ with a mild abuse of notation. Face and degeneracy maps for the simplicial set $R(e)$ are also thus given by precomposition by the appropriate maps to Δ^n (called a cosimplicial object since it is covariant). Levelwise postcomposition makes R functorial.

The geometric realization of general simplicial sets is as follows. There is a natural functor $\Delta \rightarrow \mathcal{T}\mathbf{op}$ given by $[n] \mapsto \Delta^n$ for Δ^n the standard n -simplex. It has both left- and right-adjoints. Since $\mathcal{T}\mathbf{op}$ is cocomplete and locally small, plugging in $\mathcal{T}\mathbf{op}$ for $\mathcal{E}$ in the previous discussion gives an adjunction $L \dashv R$ with $L : \mathbf{sSet} \rightarrow \mathcal{T}\mathbf{op}$ the left Kan extension, and $R : \mathcal{T}\mathbf{op} \rightarrow \mathbf{sSet}$ its right adjoint. R is called the *total singular complex functor*, denoted usually by S , and L is the *geometric realization*, denoted usually by $|-|$. It's computed as

$$|X| = \int^{n \in \Delta} X_n \times \Delta^n = \text{coeq} \left(\coprod_{f \in \text{Mor}(\Delta)} X_{\text{cod } f} \cdot \Delta^{\text{dom } f} \rightrightarrows \coprod_{[n] \in \Delta} X_n \cdot \Delta^n \right).$$

Here we note that copower is isomorphic to cartesian proeuct in $\mathcal{T}\mathbf{op}$. This gives $|y(n) = \Delta^n| = \Delta^n$, once again justifying the annoying notation.

Let $\Delta^\bullet : \Delta \rightarrow \mathbf{Cat}$ now be the functor mapping $[n]$ to the poset category $n + 1$ and maps to functors. Indeed, $\Delta^\bullet$ is a full embedding $\Delta \hookrightarrow \mathbf{Cat}$. Its right adjoint this time is the *nerve* functor $N : \mathbf{Cat} \rightarrow \mathbf{sSet}$. Recall that $N(\mathcal{C})$ is a simplicial set with an n -simplex given by a string of n composable arrows in $\mathcal{C}$, or equivalently any functor $[n] \rightarrow \mathcal{C}$. The i th degeneracy maps insert the identity arrow, and the i th face map composes i and $i + 1$, unless $i = 0$ or n , in which case the arrow is dropped.

The left adjoint $h : \mathbf{sSet} \rightarrow \mathbf{Cat}$ maps a simplicial set to its homotopy category hX . The objects are 0-simplices (also called vertexes) of X , and its arrows are freely generated on 1-simplices with respect to relations governed by 2-simplices.

Chapter 2

Derived functors via deformation

I have some familiarity with the beginning material in this section, and will read those parts rather quickly compared to the other parts.

2.1 Homotopical categories and derived functors

The ideas here were introduced around 20 years ago, but we depart slightly from the original terminology. I feel this is important to note to avoid confusions with the literature on the topic.

Definition 2.1 (Homotopical Category). A *homotopical category* is a category $\mathcal{M}$ together with a wide subcategory $\mathcal{W}$ (wide meaning containing all objects) such that for any composable triple f, g, h of arrows, if hg and gf are in $\mathcal{W}$, then $f, g, h, hgf \in \mathcal{W}$. We call arrows in $\mathcal{W}$ *weak equivalences*, and this property is called the 2-of-6 property. 

Remark. Weak equivalences in a model category satisfy the 2-of-6 property, so this construction is more general than model categories. However, the construction is not as general as the 2-of-3 property: if two of f, g, gf are in $\mathcal{W}$, then so is the third. This property required for a category to have weak equivalences, and some use this notion to be precisely the same as that of a homotopical category. We distinguish between them and choose to work in categories which satisfy the stronger 2-of-6 property.

Homotopical categories always include all isomorphisms. This is easily proven. Since $\mathcal{W}$ is a subcategory including all objects, it equivalently includes all identities. Consider inverse morphisms $f : a \rightarrow b$ and $g : b \rightarrow a$, and consider the composable triple (f, g, f) . Note $fg = 1_b \in \mathcal{W}$ and $gf = 1_a \in \mathcal{W}$, so we must have $f, g, f, fgf = f, g \in \mathcal{W}$. Conversely, every category admits a minimal collection of weak equivalences whereby it may form a homotopical category, given by the set of isomorphisms in $\mathcal{M}$. In absence of another obvious notion of weak equivalence, we assume the minimal homotopical structure.

Definition 2.2 (Homotopy Category). The homotopy category $\mathrm{Ho} \mathcal{M}$ of a homotopical category $(\mathcal{M}, \mathcal{W})$ is the formal localization of $\mathcal{M}$ at $\mathcal{W}$. 

An explicit construction can be given by taking $\mathrm{Obj}(\mathrm{Ho} \mathcal{M}) = \mathrm{Obj}(\mathcal{M})$, and morphisms to be equivalence classes of zig-zags in $\mathcal{M}^1$ with only those in $\mathcal{W}$ permitted to go backwards, quotiented by the relations

- adjacent arrows pointing in the same direction are equal to their composition;

¹Recall zig-zags are chains of morphisms not necessarily in the composable direction, meaning some are permitted to go backwards

- adjacent pairs $\xleftarrow{w} \xrightarrow{w}$ and $\xrightarrow{w} \xleftarrow{w}$ may be removed;
- identities may be removed.

This defines for each $w \in \mathcal{W}$ a formal inverse w^{-1} which behaves in the expected way. For example, with $\mathcal{W} \subseteq \mathcal{Top}$ the set of homotopy equivalences, we obtain that for $fg \sim 1$, $g \in [f^{-1}]$.

There is a canonical localization functor $\gamma : \mathcal{M} \rightarrow \mathcal{HoM}$ mapping $x \mapsto x$ and $f \mapsto [f]$, and it is universal with respect to functors which invert weak equivalences. Hence, precomposition with γ is a bijection between functors $\mathcal{M} \rightarrow \mathcal{N}$ which invert weak equivalences and functors $\mathcal{HoM} \rightarrow \mathcal{N}$.

The homotopy category of the minimal homotopical category is isomorphic to the original category. For another example, a *weak homotopy equivalence* in $\mathcal{Top}$ is a continuous function $f : X \rightarrow Y$ which is an isomorphism of sets of paths, meaning for $U : \mathcal{Grpoid} \rightarrow \mathcal{Set}$ the forgetful functor $\mathcal{G} \mapsto \mathcal{Obj}(\mathcal{G})$, we have

$$(U\pi_0)(f) : (U\pi_0)(X) \cong (U\pi_0)(Y).$$

Taking weak homotopy equivalences to be weak equivalences, the homotopy category thereof is *equivalent* to the category of CW complexes and homotopy classes of maps.

Remark. In general, not all arrows inverted by localization are weak equivalences. A category for which those arrows inverted are precisely weak equivalences is called *saturated*. Quillen showed all model categories are saturated. We can say slightly more.

Lemma 2.1. *If $\mathcal{M}$ is a category with a collection of arrows $\mathcal{W}$ with a saturated localization, then $(\mathcal{M}, \mathcal{W})$ is homotopical.*

Proof. Saturated means weak equivalences in $\mathcal{M}$ are precisely isomorphisms in $\mathcal{HoM}$. Since isomorphisms satisfy the 2-of-6 property, so must the weak equivalences. ■

Annoyingly, homotopy categories need not be locally small. Quillen proved model categories induce locally small localizations, but in the more general setting it is preferable to avoid working directly in the homotopy category to avoid these issues.

Definition 2.3 (Homotopical Functor). A homotopical functor $F : \mathcal{M} \rightarrow \mathcal{N}$ between homotopical categories is a functor which preserves weak equivalences. ♥

By univeraslity, homotopical functors induce unique functors

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{F} & \mathcal{N} \\ \gamma \downarrow & & \downarrow \delta \\ \mathcal{HoM} & \xrightarrow{F} & \mathcal{HoN} \end{array}$$

Since we often without loss of generality call both functors F , we say that F commutes with the localizations.

Remark. Let $\mathcal{N}$ be a minimal homotopic category. Universality of localization $\gamma : \mathcal{M} \rightarrow \mathcal{HoM}$ here is 2-categorical: natural transformations between *homotopical* functors $\mathcal{M} \rightarrow \mathcal{N}$ are bijective with natural transformations $\mathcal{HoM} \rightarrow \mathcal{N}$. If $\mathcal{N}$ is not minimal, we must distinguish between $\mathcal{HoN}$ and $\mathcal{N}$. For δ the localization functor, each natural transformation α between functors $\mathcal{M} \rightarrow \mathcal{N}$ lowers uniquely to a natural transformation $\delta\alpha$ between functors $\mathcal{HoM} \rightarrow \mathcal{HoN}$ (since $\mathcal{HoN}$ inverts weak

equivalences and our functors are homotopical). Although this relation is unique, it is not necessarily bijective, and transformations between functors $\text{Ho}\mathcal{M} \rightarrow \text{Ho}\mathcal{N}$ can always raise to one between functors $\mathcal{M} \rightarrow \text{Ho}\mathcal{N}$, but we may not be able to lift it along δ .

Example. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a (not necessarily exact) additive functor between abelian categories. Denote by $\text{Ch}_{\geq 0}(\mathcal{A})$ and $\text{Ch}_{\geq 0}(\mathcal{B})$ the categories of chain complexes bounded below at 0. The induced functor $F_{\bullet} : \text{Ch}_{\geq 0}(\mathcal{A}) \rightarrow \text{Ch}_{\geq 0}(\mathcal{B})$ is then homotopical if we take weak equivalences to be *chain complex homotopy equivalences*, but not necessarily if we take them to be quasi-isomorphisms. For example, consider a quasi-isomorphism $f_{\bullet} : A_{\bullet} \rightarrow B_{\bullet}$ with $\mathcal{A} = \mathcal{B} = \mathcal{A}b$, given by

$$\begin{array}{ccccccc} \dots & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \xrightarrow{\sim} & \mathbb{Z}/4\mathbb{Z} & \xrightarrow{0} & \mathbb{Z}/2\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/4\mathbb{Z} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0 \end{array}$$

This is shown to be a quasi-isomorphism by computing the cohomologies. The computation is not difficult, but takes enough space to type out to make it annoying. We can apply the hom-functor $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, -)$, and we compute

$$\begin{aligned} \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, 0) &= 0, & \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) &= \{0, 1\} = \mathbb{Z}/2\mathbb{Z}, \\ \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}) &= \{0, (0, 1) \mapsto (0, 2)\} = \mathbb{Z}/2\mathbb{Z}. \end{aligned}$$

Hence we obtain

$$\begin{array}{ccccccc} \dots & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \xrightarrow{\cong} & \mathbb{Z}/2\mathbb{Z} & \xrightarrow{0} & \mathbb{Z}/2\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} 0 \\ & & \downarrow & & \downarrow 0 & & \downarrow 0 \\ \dots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0. \end{array}$$

The top row is exact, but note that the homology of the bottom complex at any $\mathbb{Z}/2\mathbb{Z}$ is

$$\frac{\text{Ker}(\mathbb{Z}/2\mathbb{Z} \rightarrow 0)}{\text{Im}(0 \rightarrow \mathbb{Z}/2\mathbb{Z})} = \frac{\mathbb{Z}/2\mathbb{Z}}{0} = \mathbb{Z}/2\mathbb{Z} \neq 0.$$

Hence, $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, f)_{\bullet}$ is not a quasi-isomorphism, and additive functors need not be homotopic for this class of weak equivalences.

Many other examples show there are interesting non-homotopical functors. One such class is that of *derived functors*, which are in some sense the closest approximation to homotopical.

Definition 2.4 (Total Derived Functor). A *total left derived functor* $\mathbf{L}F$ of some functor $F : \mathcal{M} \rightarrow \mathcal{N}$ for $\mathcal{M}, \mathcal{N}$ homotopic is a *right* Kan extension $\text{Ran}_{\gamma} \delta F$

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{F} & \mathcal{N} \\ \gamma \downarrow & \uparrow & \downarrow \delta \\ \text{Ho}\mathcal{M} & \xrightarrow[\mathbf{L}F]{} & \text{Ho}\mathcal{N} \end{array}$$

with γ, δ localization functors. Dually, a *right* derived functor $\mathbf{R}F$ is a *left* Kan extension $\text{Lan}_{\gamma} \delta F$. ♡

Universality tells us that total derived functors are equivalently homotopical functors $\mathcal{M} \rightarrow \text{Ho } \mathcal{N}$, which are often called “derived functors”. We reserve that terminology for a special case in which this correspondence is a lift along δ . Sometimes these are called point-set derived functors.

Definition 2.5 (Derived Functor). A *left derived functor* of $F : \mathcal{M} \rightarrow \mathcal{N}$ is a homotopical functor $\mathbb{L}F : \mathcal{M} \rightarrow \mathcal{N}$ and a natural transformation $\lambda : \mathbb{L}F \Rightarrow F$ such that $\delta \cdot \lambda : \delta \circ \mathbb{L}F \rightarrow \delta \circ F$ is a total left derived functor of F (in the sense that it is equal to the Kan extension precomposed with γ). Right derived functors are defined the same. ♥

2.2 Derived functors via deformations

Definition 2.6 (Deformation). A *left deformation* on a homotopical category $\mathcal{M}$ consists of an endofunctor Q and a natural weak equivalence $q : Q \Rightarrow 1_{\mathcal{M}}$. Dually, a right deformation is a natural weak equivalence $q : 1_{\mathcal{M}} \Rightarrow Q$. ♥

Here, that q is a natural weak equivalence means that each component q_m for $m \in \mathcal{M}$ is itself a weak equivalence. Deformations axiomatize a convenient setting in which derived functors exist, and are constructed rather easily. Necessarily Q is homotopical. This is easily shown. Let $w : a \rightarrow b$ be a weak equivalence. We then have that $q_b \circ Q(w) = w \circ q_a$. Note that since w and q_a are weak equivalences, their composition must also be a weak equivalence by the 2-of-3 property, which is implied by the 2-of-6 property. Hence, $q_b \circ Q(w)$ is a weak equivalence, and again by the 2-of-3 property we have that $Q(w)$ is a weak equivalence. Now let $\mathcal{M}_Q$ be any full subcategory containing the image of Q . The inclusion $\mathcal{M}_Q \hookrightarrow \mathcal{M}$ and Q define an equivalence of categories between $\text{Ho } \mathcal{M}$ and $\text{Ho } \mathcal{M}_Q$. The book calls $\mathcal{M}_Q$ the subcategory of *cofibrant objects*, and specifies they use the term to mean something different than Quillen’s definition in model categories. I will try to abide by another convention and call it a *left deformation retract*.

Definition 2.7 (Deformation for Functors). A *left deformation* for a functor $F : \mathcal{M} \rightarrow \mathcal{N}$ between homotopical categories is a left deformation Q for $\mathcal{M}$ such that F is homotopical on a left deformation retract $\mathcal{M}_Q \subseteq \mathcal{M}$. We say F is *left deformable* when it admits a left deformation. ♥

In the context of model categories, Ken Brown’s lemma shows that a left Quillen functor (a cocontinuous² functor preserving cofibrations and trivial cofibrations) will preserve weak equivalences between cofibrant objects (objects where the unique morphism from the initial object are cofibrations). We prove this lemma later in the text.

As we continue, when we specify a left deformation for a category $\mathcal{M}$, we implicitly choose a left deformation retract.

Theorem 2.1. *If $F : \mathcal{M} \rightarrow \mathcal{N}$ has a left deformation $q : Q \Rightarrow 1_{\mathcal{M}}$, then $\mathbb{L}F = F \circ Q$ is a left derived functor of F .*

Proof. Let $\delta : \mathcal{N} \rightarrow \text{Ho } \mathcal{N}$ the localization. By definition 2.5, we must show δFQ and $\delta Fq : \delta FQ \Rightarrow \delta F$ form a left Kan extension of δF along γ . Equivalently, we show it satisfies the relevant universal property in $\text{Fun}(\mathcal{M}, \text{Ho } \mathcal{N})$. Let $H : \mathcal{M} \rightarrow \text{Ho } \mathcal{N}$ be homotopical (hence meaning it inverts weak equivalences), and $\alpha : G \Rightarrow \delta F$ the relevant natural transformation. Since G is homotopical and q is a natural weak equivalence, $Gq : GQ \Rightarrow G$ is a natural isomorphism. Naturality of α

²Conventions appear to vary, this book uses cocontinuous and says some texts require only that (trivial) cofibrations be preserved, but other sources I can find require that it be a left adjoint functor, strengthening the hypothesis.

gives that

$$\delta Fq \circ \alpha Q \circ (Gq)^{-1} = (\delta F \circ \alpha)(q \circ Q) \circ (Gq)^{-1} = \alpha q \circ (Gq)^{-1} = \alpha.$$

Hence, α factors through δFq . To prove uniqueness, first note that since Q is homotopical, qQ is a natural weak equivalence. Since Q is a deformation of F , we have that FqQ is a natural weak equivalence, and thus δFqQ is a natural isomorphism. Let α' be another factorization of α through δFq . We then obtain

$$\begin{array}{ccc} GQ & \xrightarrow{\alpha'Q} & \delta FQ^2 \\ Gq \downarrow & & \downarrow \delta FQq \\ G & \xrightarrow{\alpha'} & \delta FQ \end{array}$$

This indeed commutes since Gq is still invertible, so flipping that arrow gives the same equality as above. This indeed also shows the required uniqueness. $\blacksquare$

Hence for $r : 1 \Rightarrow R$ a right deformation and $\mathcal{M}_R$ a right deformation retract, we have also that $\mathbb{R}F = FR$.

Denote by $\mathcal{L}\text{Def}$ the 2-category of homotopical categories equipped with specified left deformations and left deformation retracts, morphisms given by functors that restrict to homotopical functors on the deformation retracts (left deformable functors with respect to the specified deformation), and 2-morphisms as natural transformations. Note here that a *pseudofunctor* is a functor $\mathcal{Z} \rightarrow \text{Cat}$ which only preserves identities and composition up to coherent natural isomorphism.

Theorem 2.2. *There is a pseudofunctor $\mathbf{L} : \mathcal{L}\text{Def} \rightarrow \text{Cat}$ mapping homotopical categories to their homotopy category, left deformable functors to their total left derived functor, and natural transformations to the derived natural transformations.*

The proof of this theorem (which I omit :3) depends on our construction in theorem 2.1, and it does not hold in generality that the composite of total derived functors is total derived for the composite.

Definition 2.8 (Deformable Adjunction). An adjunction $F \dashv G$ is a *deformable adjunction* if the left adjunct F is left deformable, and the right adjunct G is right deformable. $\heartsuit$

Theorem 2.3. *If*

$$\mathcal{M} \begin{array}{c} \xrightarrow{F} \\ \perp \\ \xleftarrow{G} \end{array} \mathcal{N}$$

is a deformable adjunction, then so is

$$\mathbf{L}F : \quad \text{Ho } \mathcal{M} \begin{array}{c} \xrightarrow{\quad} \\ \perp \\ \xleftarrow{\quad} \end{array} \text{Ho } \mathcal{N} \quad : \mathbf{R}G.$$

Furthermore, the total derived adjunction $\mathbf{L}F \dashv \mathbf{R}G$ is the unique adjunction compatible with the adjunctions in the sense that

$$\begin{array}{ccc} \mathcal{N}(F(m), n) & \cong & \mathcal{M}(m, G(n)) \\ \delta \downarrow & & \downarrow \gamma \\ \text{Ho } \mathcal{M}(F(m), n) & & \text{Ho } \mathcal{N}(m, G(n)) \\ Fq^* \downarrow & & \downarrow Gr_* \\ \text{Ho } \mathcal{N}(\mathbf{L}F(m), n) & \cong & \text{Ho } \mathcal{M}(m, \mathbf{R}G(n)) \end{array}$$

commutes for each $m \in \mathcal{M}$ and $n \in \mathcal{N}$.

The proofs of this fact are very complex, and was omitted from the text. I am not going to go chase after a proof. Instead, I choose to make sense of the statement itself. The only point of confusion is the precomposition by Fq and postcomposition by Gr , but indeed by theorem 2.2 this gives the total derived functor.

Proposition 2.1. *The total left derived functor of a left deformable functor is an absolute Kan extension; in particular, a pointwise right Kan extension.*

Proof. Let $(\delta FQ, \delta Fq)$ be the total left derived functor given by the deformation (Q, q) and theorem 2.2. Let $H : \text{Ho } \mathcal{N} \rightarrow \mathcal{E}$. We must show that H preserves the Kan extension $\mathbf{L}F$. To use our construction of $\mathbf{L}F$, we must transfer the universal property to $\mathcal{E}^{\mathcal{M}}$. Let $G : \mathcal{M} \rightarrow \mathcal{E}$ homotopical and $\alpha : G \Rightarrow H\delta F$. It suffices to show it factors uniquely through $H\delta Fq$. Similarly to the last proof, α factors as the composition

$$G \xrightarrow{(Gq)^{-1}} GQ \xrightarrow{\alpha Q} H\delta FQ \xrightarrow{H\delta Fq} H\delta F.$$

By the same logic as the above proof, this factorization is unique as well. ■

Exercise. Let $F \dashv G$ and let $\mathbf{L}F$ and $\mathbf{R}G$ be absolute Kan extensions. Show that $\mathbf{L}F \dashv \mathbf{R}G$.

Proof. Let $\eta : 1_{\mathcal{M}} \Rightarrow GF$ be the unit and $\varepsilon : FG \Rightarrow 1_{\mathcal{N}}$ the counit of the adjunction $F \dashv G$. Let $\gamma : \mathcal{M} \rightarrow \text{Ho } \mathcal{M}$ and $\delta : \mathcal{N} \rightarrow \text{Ho } \mathcal{N}$ the localization functors. Let $\beta : \delta G \Rightarrow \mathbf{R}G$ and $\alpha : \mathbf{L}F \Rightarrow \delta F$ the derived natural transformations. ■

A double category Der with objects *deformable categories* (homotopical categories with both left and right deformations, such that either R preserves $\mathcal{M}_Q$ or vice versa), horizontal 1-cells right deformable functors preserving fibrant objects, vertical 1-cells left deformable functors preserving cofibrant objects, and 2-cells appropriate natural transformations. There is then a double pseudofunctor $\mathcal{D} \rightarrow \text{Cat}$ given by $\mathcal{C} \mapsto \text{Ho } \mathcal{C}$ and $F \mapsto \mathbf{L}F$; here Cat is the double category of categories, functors, and natural transformations. No way in hell I'm proving that.

2.3 Classical derived functors between abelian categories

Recall that Mod_R has enough projectives, meaning for any A there is a bounded below complex of projectives $P_{\bullet}$ quasi-isomorphic to $\iota(A)$ (a projective resolution of A). Careful choice of projective resolutions is functorial $Q : \text{Ch}_{\geq 0}(R) \rightarrow \text{Ch}_{\geq 0}(R)$ with a natural quasi-isomorphism $q : Q \Rightarrow 1$. Dually, modules admit injective resolutions which give a right deformation $r : 1 \Rightarrow (R : \text{Ch}_{\geq 0}(R) \rightarrow \text{Ch}_{\geq 0}(R))$ on the category of bounded below cochain complexes.

Additive functors $F : \text{Mod}_R \rightarrow \text{Mod}_S$ induce functors $F_{\bullet} : \text{Ch}_{\geq 0}(R) \rightarrow \text{Ch}_{\geq 0}(S)$ and $F^{\bullet} : \text{Ch}_{\leq 0}(R) \rightarrow \text{Ch}_{\leq 0}(S)$ preserving chain homotopies. When working with projective/injective resolutions, the quasi-isomorphisms become homotopy equivalences, and are thus preserved by $F_{\bullet}$ and $F^{\bullet}$. Hence, with respect to quasi-isomorphisms, $F^{\bullet}$ has a right derived functor and $F_{\bullet}$ has a left derived functor, as described herein. The functor that we classically refer to as the left derived functor of $F : \text{Mod}_R \rightarrow \text{Mod}_S$ in homological algebra is the composition

$$\text{Mod}_R \xrightarrow{\iota} \text{Ch}_{\geq 0}(R) \xrightarrow{\mathbb{L}F_{\bullet}} \text{Ch}_{\geq 0}(S) \xrightarrow{H_0} \text{Mod}_S.$$

One can also take the other derived functors, but the derivations we chose induce long exact sequences,

so we prefer those.

2.4 Preview of homotopy limits and colimits

Let $\mathcal{M}$ a complete and cocomplete homotopical category, and $\mathcal{D}$ a category indexing some small diagram. The colimit and limit functors $\varinjlim, \varprojlim : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}$ need not be homotopical with respect to pointwise weak equivalences in $\mathcal{M}^{\mathcal{D}}$ (I assume meaning natural weak equivalences). However, since the functors are continuous, their respective Kan extensions exist, so we would want to define a left derived functor of $\varinjlim$ and a right derived functor of $\varprojlim$, called hocolim and holim, respectively. Unfortunately, even if $\mathcal{M}$ is a model category, $\mathcal{M}^{\mathcal{D}}$ may not have a model structure such that $\varinjlim$ and $\varprojlim$ are Quillen. Hence the theory of deformations exists for construction derived functors without invoking model structures.

